



网络信息处理实验报告

实验 4：路由配置实验

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一、实验名称

路由配置实验

二、实验任务

1.实验一：静态路由实验

根据实验拓扑图，完成如下任务：

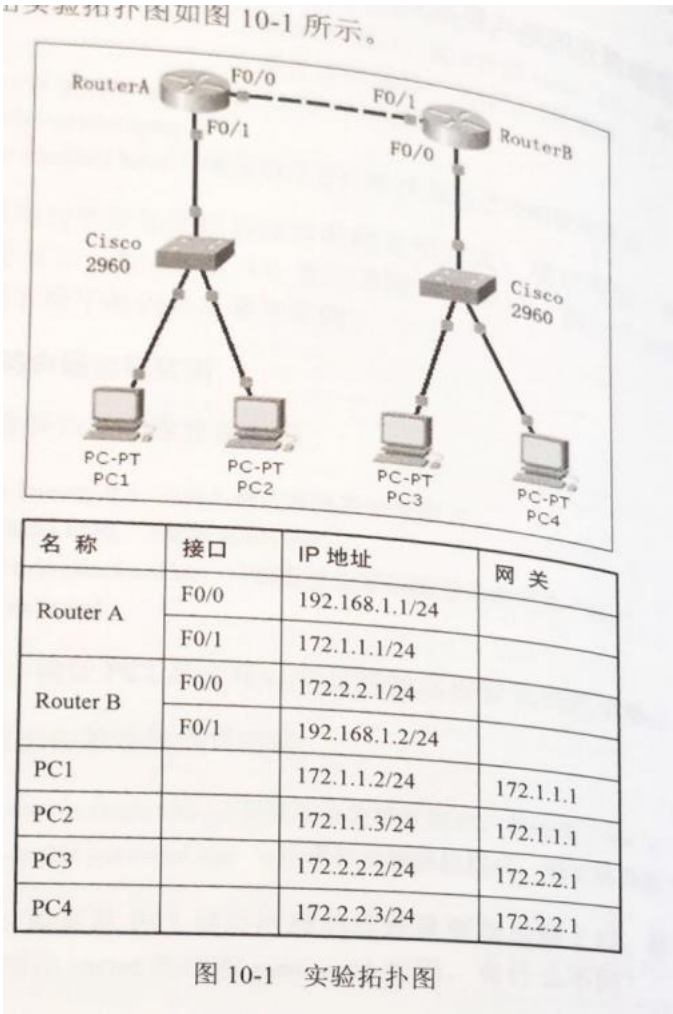


Figure 1 实验拓扑图

- (1) 路由器的基本配置：关闭域名解释；设置路由器接口 IP 地址。
- (2) 根据以上拓扑划分出的三个网段，要求配置静态路由以达到所有客户机都能相互通信。
- (3) 配置默认路由。
- (4) 了解 Ping 命令和 trace（跟踪）的原理和使用。

2.实验二：路由信息协议（RIP）实验

根据实验拓扑图（同图 2-1），完成如下任务：

- (1) 路由器的基本配置：设置路由器接口 IP 地址。
- (2) 根据以上拓扑划分出的三个网段，要求配置 RIP 路由使所有客户机都能相互通信，该如何实现？

3.实验三：开放最短路径有限（OSPF）实验

根据实验拓扑图（同图 2-1），完成如下任务：

- （1）路由器的基本配置：设置路由器接口 IP 地址。
- （2）根据以上拓扑划分出的 3 个网段，要求配置 OSPF 路由使所有客户机都能相互通信。

三、实验环境及工具

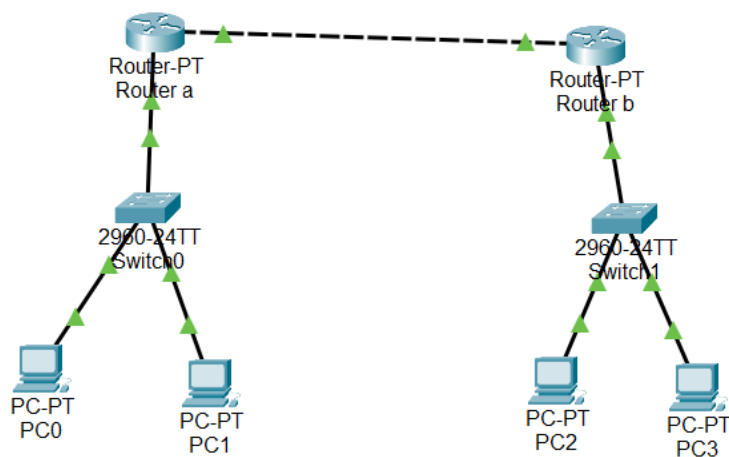
本实验使用 Cisco Packet Tracer8.2 版本。

四、实验记录与结果分析

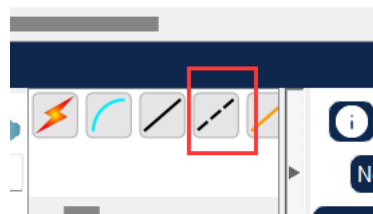
1. 任务 1：

（1）路由器的基本配置：

根据实验步骤，配置 IP 信息等：



连线部分需要注意，使用下图所示的双绞线：



来连接两个 router，并且连接的接口要是全球 ip 地址所在的接口。以 router a 为例，

(1) Router A 的基本配置如图 10-2 所示。

```
Router>en
Router#config t
Router(config)#no ip domain-look (关闭域名解释)
Router(config-if)#int f0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown (默认是shutdown)
Router(config)#exit
Router(config)#int f0/1
Router(config-if)#ip address 172.1.1.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#end
```

图 10-2 Router A 的基本配置

```
Router5
Physical Config Attributes
IOS Command Line Interface

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
% Invalid input detected at '^' marker.

Router(config)#int f0/0
Router(config-if)#exit
Router(config)#int f0/1
%Invalid interface type and number
Router(config)#int f0/1
%Invalid interface type and number
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/1
%Invalid interface type and number
Router(config)#no ip domain-look
Router(config)#int f0/1
%Invalid interface type and number
Router(config)#int f1/0
Router(config-if)#ip address 172.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

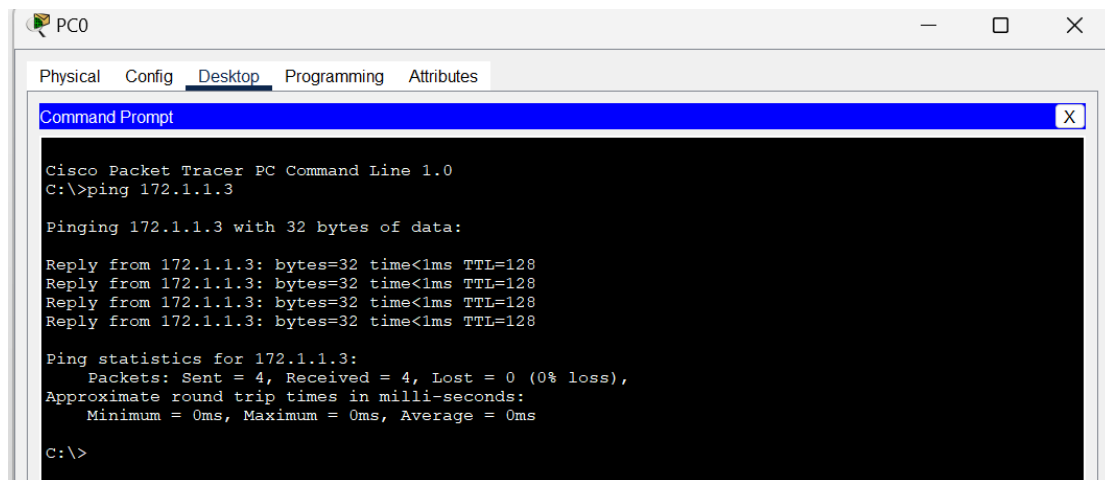
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

在控制台中输入相应的配置信息。

(2) PC1 上分别 Ping 主机 PC2 以及路由器 B 测试通信情况记录 Ping 命令的响应信息。思考:为什么会产生这样的信息?

从 PC0 ping PC1



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.1.1.3

Pinging 172.1.1.3 with 32 bytes of data:

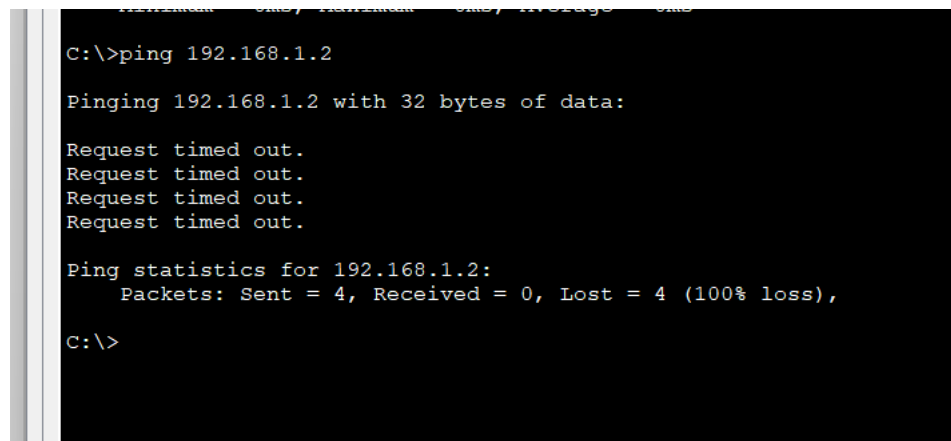
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.1.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

二者之间是连通的。

从 pc0 ping 路由器 B:



```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

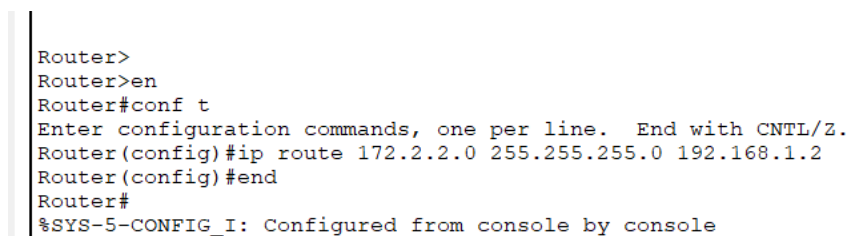
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

二者之间是连接失败。

(3) 配置静态路由:

配置 router a:



```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.2.2.0 255.255.255.0 192.168.1.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

(4) 再次对 pc2 和路由器 b 进行 ping 操作:

仅配置 router a:

```
C:\>ping 172.1.1.3

Pinging 172.1.1.3 with 32 bytes of data:

Reply from 172.1.1.3: bytes=32 time<1ms TTL=128
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128
Reply from 172.1.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.1.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
```

在配置了 Router0 的静态路由的条件下，因为没有配置 Router1 的静态路由，信息可以从 PC0 到 Router1，但 Router1 无法进行反馈。

将 router b 也配置上：

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=9ms TTL=254
Reply from 192.168.1.2: bytes=32 time<1ms TTL=254
Reply from 192.168.1.2: bytes=32 time<1ms TTL=254
Reply from 192.168.1.2: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

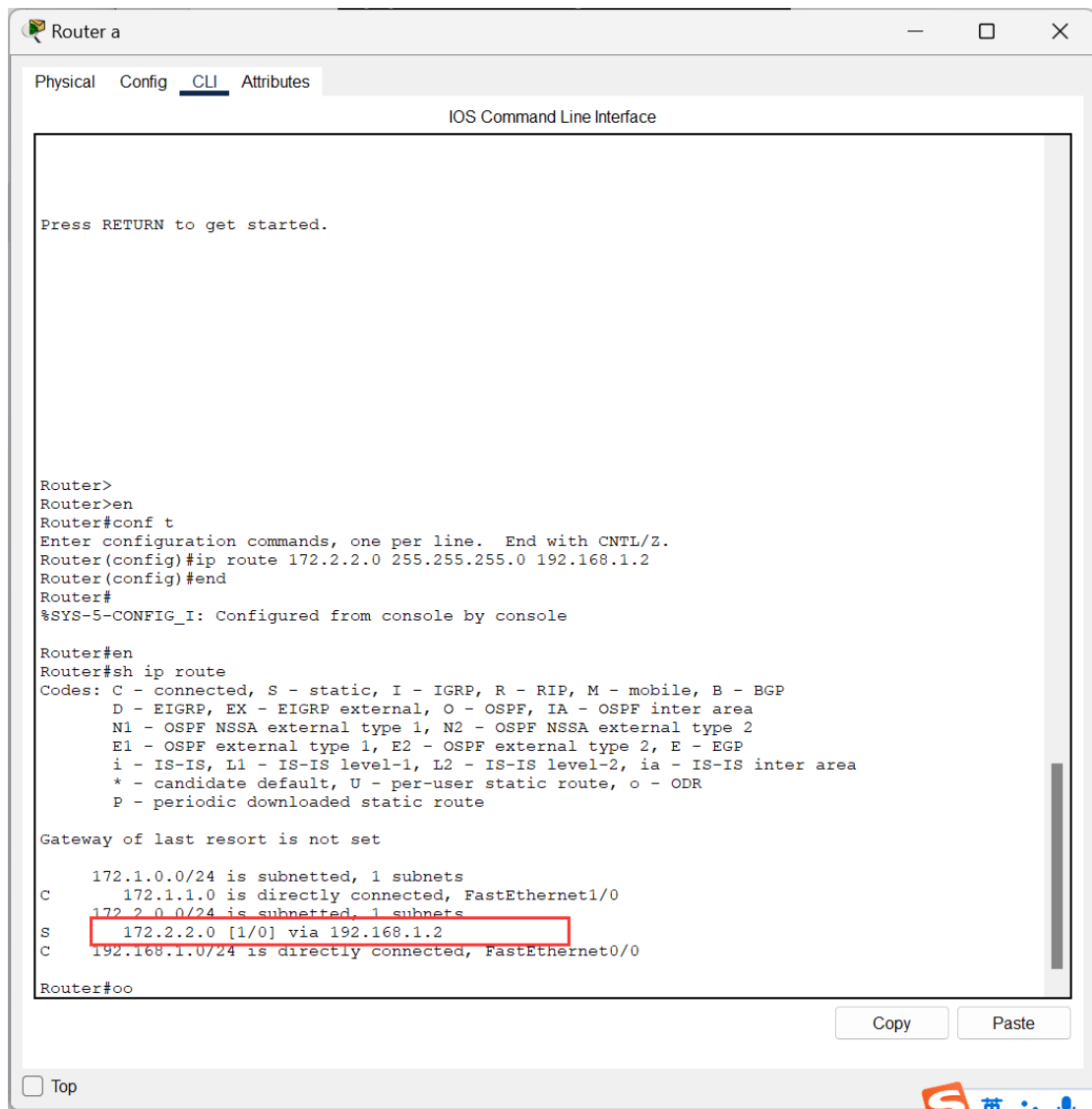
C:\>
```

[\] Top](#)

可以收到来自 router b 的回复。

(5) 查看配置：

Router a 的配置信息：



Router b 的配置信息：

```
Router#en
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.1.0.0/24 is subnetted, 1 subnets
S      172.1.1.0 [1/0] via 192.168.1.1
C      172.2.0.0/24 is subnetted, 1 subnets
C      172.2.2.0 is directly connected, FastEthernet0/0
C      192.168.1.0/24 is directly connected, FastEthernet1/0

Router#
```

静态路由信息都配置成功。

(6) ping 测试：

从 pc0 到 pc2 和 3 都成功,

```
C:\>ping 172.2.2.2

Pinging 172.2.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time=1ms TTL=126
Reply from 172.2.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.2.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

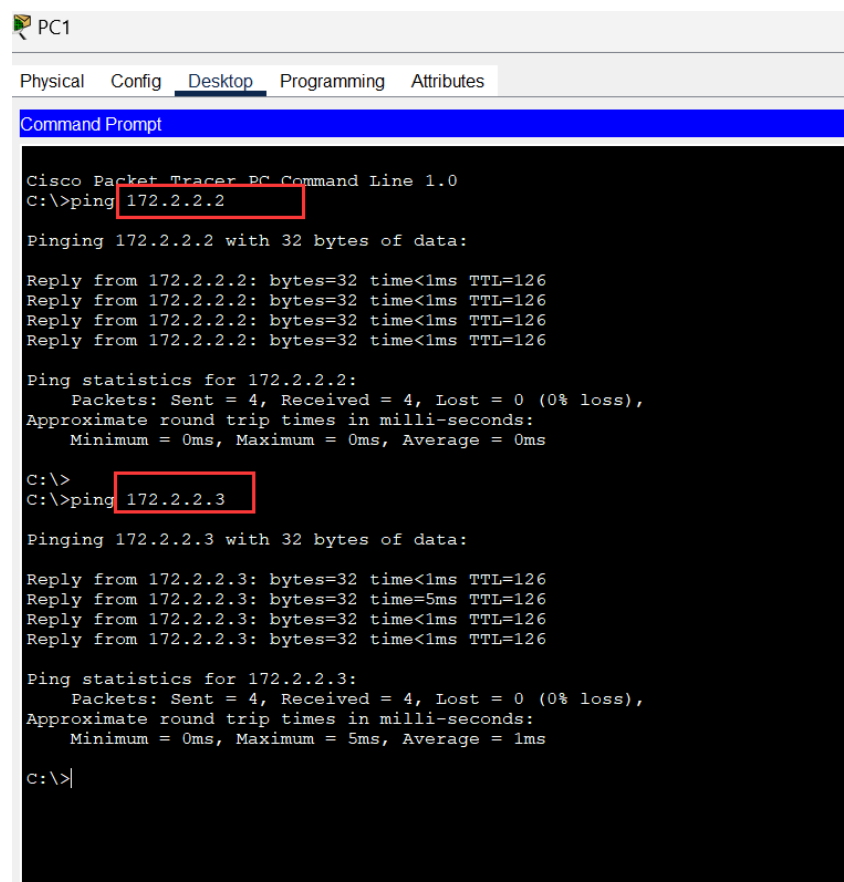
C:\>ping 172.2.2.3

Pinging 172.2.2.3 with 32 bytes of data:

Request timed out.
Reply from 172.2.2.3: bytes=32 time<1ms TTL=126
Reply from 172.2.2.3: bytes=32 time<1ms TTL=126
Reply from 172.2.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 172.2.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

从 pc1 到 pc2 和 3 都成功,



The screenshot shows the Command Prompt window of PC1 in Cisco Packet Tracer. The window title is 'PC1' and it has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The command prompt displays the output of two ping commands: 'ping 172.2.2.2' and 'ping 172.2.2.3'. Both commands show successful results with 0% loss. The IP addresses 172.2.2.2 and 172.2.2.3 are highlighted with red boxes in the original image.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.2.2.2

Pinging 172.2.2.2 with 32 bytes of data:

Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.2.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>ping 172.2.2.3

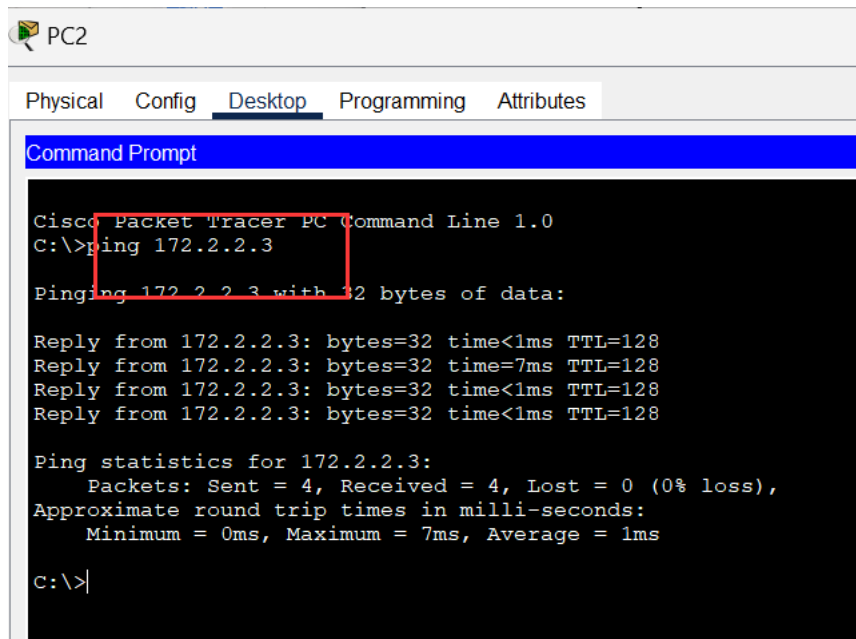
Pinging 172.2.2.3 with 32 bytes of data:

Reply from 172.2.2.3: bytes=32 time<1ms TTL=126
Reply from 172.2.2.3: bytes=32 time=5ms TTL=126
Reply from 172.2.2.3: bytes=32 time<1ms TTL=126
Reply from 172.2.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 172.2.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>|
```

从 pc2 到 pc3 成功,



The screenshot shows the 'Desktop' tab of a PC2 in Cisco Packet Tracer. A Command Prompt window is open, displaying the output of a ping command to 172.2.2.3. The command 'ping 172.2.2.3' is highlighted with a red box. The output shows four successful replies with 32 bytes of data, times less than 1ms, and a TTL of 128. Ping statistics indicate 4 packets sent, 4 received, and 0% loss, with an average round trip time of 1ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.2.2.3

Pinging 172.2.2.3 with 32 bytes of data:

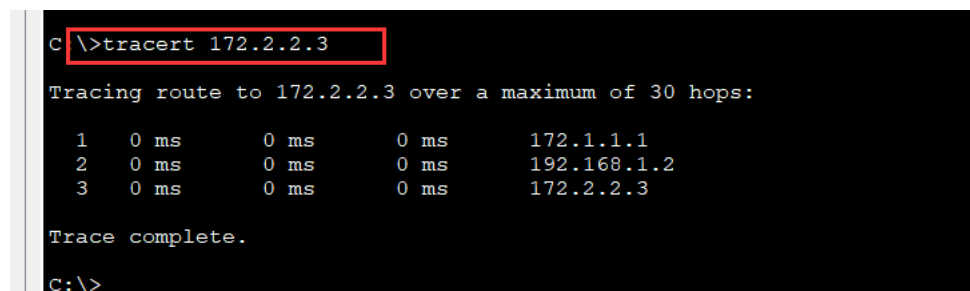
Reply from 172.2.2.3: bytes=32 time<1ms TTL=128
Reply from 172.2.2.3: bytes=32 time=7ms TTL=128
Reply from 172.2.2.3: bytes=32 time<1ms TTL=128
Reply from 172.2.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.2.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>
```

所有主机节点之间均是相互连通。

(7) 追踪路由：



The screenshot shows a Command Prompt window with the output of the 'tracert 172.2.2.3' command. The command is highlighted with a red box. The output shows the route from the source to the destination, with three hops. Each hop shows the IP address of the router and the time taken for the packet to reach it. The trace is complete.

```
C:\>tracert 172.2.2.3

Tracing route to 172.2.2.3 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    172.1.1.1
  1  0 ms    0 ms    0 ms    192.168.1.2
  2  0 ms    0 ms    0 ms    172.2.2.3

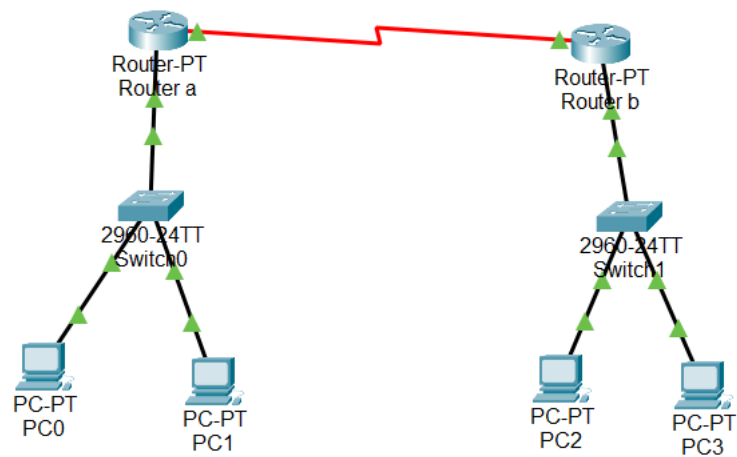
Trace complete.

C:\>
```

在 PC0 中，使用 tracert 命令，可以查看从 PC0 到 PC3 经过的路由器和花费的时间。

思考题：

(1) 基本配置：



Router a 是 DCE:

```

Router#show controllers s2/0
Interface Serial2/0
Hardware is PowerQUICC MPC860
DCE V.35, no clock
ldb at 0x81081AC4, driver data structure at 0x81084AC0
SCC Registers:
General [GSMR]=0x2:0x00000000, Protocol-specific [PSMR]=0x8
Events [SCCE]=0x0000, Mask [SCCM]=0x0000, Status [SCCS]=0x00
Transmit on Demand [TODR]=0x0, Data Sync [DSR]=0x7E7E
Interrupt Registers:
Config [CICR]=0x00367F80, Pending [CIPR]=0x0000C000
Mask [CIMR]=0x00200000, In-srv [CISR]=0x00000000
Command register [CR]=0x580
Port A [PADIR]=0x1030, [PAPAR]=0xFFFF
[PAODR]=0x0010, [PADAT]=0xCBFF
Port B [PBDIR]=0x09C0F, [PBPAR]=0x0800E
[PBODR]=0x00000, [PBDAT]=0x3FFFD
Port C [PCDIR]=0x00C, [PCPAR]=0x200
[PCSO]=0xC20, [PCDAT]=0xDF2, [PCINT]=0x00F
Receive Ring
  rmd(68012830): status 9000 length 60C address 3B6DAC4
  rmd(68012838): status B000 length 60C address 3B6D444
Transmit Ring
--More--
  
```

Router b 是 DTE:

```
Router#show controllers s2/0
Interface Serial2/0
Hardware is PowerQUICC MPC860
DTE V.35 TX and RX clocks detected
idb at 0x81081AC4, driver data structure at 0x81084AC0
SCC Registers:
General [GSMR]=0x2:0x00000000, Protocol-specific [PSMR]=0x8
Events [SCCE]=0x0000, Mask [SCCM]=0x0000, Status [SCCS]=0x00
Transmit on Demand [TODR]=0x0, Data Sync [DSR]=0x7E7E
Interrupt Registers:
Config [CICR]=0x00367F80, Pending [CIPR]=0x0000C000
Mask [CIMR]=0x00200000, In-srv [CISR]=0x00000000
Command register [CR]=0x580
Port A [PADIR]=0x1030, [PAPAR]=0xFFFF
[PAODR]=0x0010, [PADAT]=0xCBFF
Port B [PBDIR]=0x09C0F, [PBPAR]=0x0800E
[PBODR]=0x000000, [PBDAT]=0x3FFFD
Port C [PCDIR]=0x00C, [PCPAR]=0x200
[PCSO]=0xC20, [PCDAT]=0xDF2, [PCINT]=0x00F
Receive Ring
  rmd(68012830): status 9000 length 60C address 3B6DAC4
  rmd(68012838): status B000 length 60C address 3B6D444
Transmit Ring
--More--
```

Copy

Top

Time: 01:16:49

再给 DCE 配置一个时钟频率：

```
Router(config)#int s2/0
Router(config-if)#clock rate 64000
Router(config-if)#
```

然后用前面使用的静态路由配置方式进行配置。

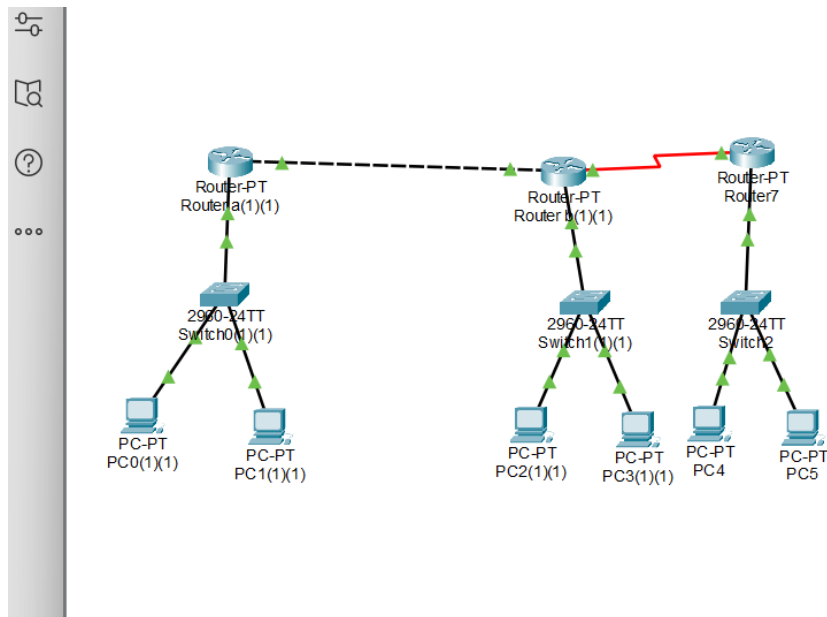
用 pc0 ping pc2:

```
:>ping 172.2.2.2
inging 172.2.2.2 with 32 bytes of data:
equest timed out.
eply from 172.2.2.2: bytes=32 time<1ms TTL=126
eply from 172.2.2.2: bytes=32 time<1ms TTL=126
eply from 172.2.2.2: bytes=32 time<1ms TTL=126
ing statistics for 172.2.2.2:
  Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
```

经过最终检验，任意两台电脑之间的 ping 都成功因此网络连通。

(2):

基本配置：



我给新添加的网络 pc4 设为 ip: 172.3.3.2 网关: 172.3.3.1; pc5 ip:172.3.3.3 网关: 172.3.3.1 ; router 7 的 f0/0 ip 172.3.3.1; se2/0 192.168.1.3; router b 的 se2/0 设为 192.168.2.1。按照实验 1 的方法对其进行配置即可。

从 pc0 ping pc4 和 5 均成功:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.3.3.2
Pinging 172.3.3.2 with 32 bytes of data:

Reply from 172.1.1.1: Destination host unreachable.
Reply from 172.1.1.1: Destination host unreachable.
Reply from 172.1.1.1: Destination host unreachable.
Reply from 172.1.1.1: Destination host unreachable.

Ping statistics for 172.3.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.3.3.3
Pinging 172.3.3.3 with 32 bytes of data:

Reply from 172.1.1.1: Destination host unreachable.
Request timed out.
Reply from 172.1.1.1: Destination host unreachable.
Reply from 172.1.1.1: Destination host unreachable.

Ping statistics for 172.3.3.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
  
```

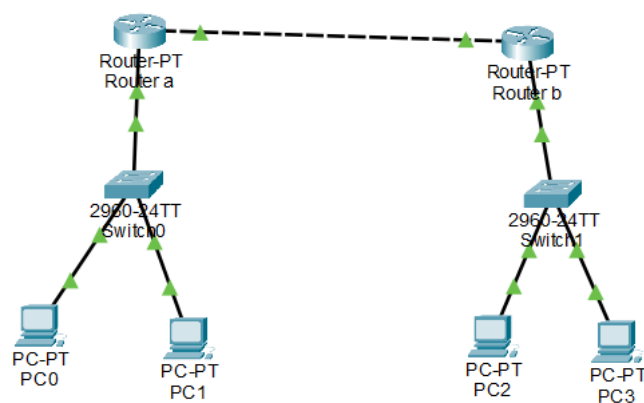
从 pc4 ping pc0 和 1 也都成功:

```
PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.1.1.2
Pinging 172.1.1.2 with 32 bytes of data:
Reply from 172.3.3.1: Destination host unreachable.
Reply from 172.3.3.1: Destination host unreachable.
Request timed out.
Reply from 172.3.3.1: Destination host unreachable.
Ping statistics for 172.1.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 172.1.1.3
Pinging 172.1.1.3 with 32 bytes of data:
Reply from 172.3.3.1: Destination host unreachable.
Request timed out.
Reply from 172.3.3.1: Destination host unreachable.
Reply from 172.3.3.1: Destination host unreachable.
Ping statistics for 172.1.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

经过检验其他 ping 同样成功，任意两台电脑之间均是连通的。

2. 任务 2：.路由信息协议（RIP）实验

(1) 同任务一中的基本配置：



(2) RIP 路由器配置：

对 router a 进行 rip 配置：

```

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#172.1.1.0
      ^
% Invalid input detected at '^' marker.

Router(config-router)#network 172.1.1.0
Router(config-router)#network 192.168.1.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#

```

对 router b 进行 rip 配置：

```

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 172.2.2.0
Router(config-router)#network 192.168.1.0
      ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.1.0
Router(config-router)#

```

☐ Top

(3) 使用 show ip route 查看信息

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    171.1.0.0/24 is subnetted, 1 subnets
C       171.1.1.0 is directly connected, FastEthernet1/0
R       172.2.0.0/16 [120/1] via 192.168.1.2, 00:00:06, FastEthernet0/0
C       192.168.1.0/24 is directly connected, FastEthernet0/0

Router#

```

Copy Pa

(4) 测试 ping:

从 p0 到 p2 测试成功

```
C:\>ping 172.2.2.2

Pinging 172.2.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.2.2.2: bytes=32 time=1ms TTL=126
Reply from 172.2.2.2: bytes=32 time=1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.2.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.2.2.2

Pinging 172.2.2.2 with 32 bytes of data:

Reply from 172.2.2.2: bytes=32 time=2ms TTL=126
Reply from 172.2.2.2: bytes=32 time=1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.2.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

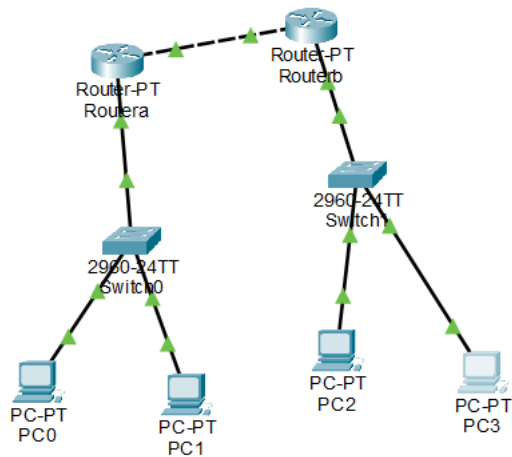
经过检验任意俩主机之间均连通。

思考题：

本题做法结合了任务1思考题的思路和任务2对RIP的配置。配置串口之后，在配置两个路由器的RIP即可。

3. 任务3：

(1) 基本配置：



(2) 配置 ospf

配置 router a:

```

Router>en
Router#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 100
Router(config-router)#net 172.1.1.0 0.0.0.255 area 0
Router(config-router)#net 192.168.1.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

配置 b:

```

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 100
Router(config-router)#net 172.2.2.0 0.0.0.255 area 0
Router(config-router)#net 192.168.1.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:40:02: %OSPF-5-ADJCHG: Process 100, Nbr 192.168.1.1 on FastEthernet1/0 from LOADING to FULL,
loading Done

```

(3) 查看路由表项:


```
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

171.1.0.0/24 is subnetted, 1 subnets
C    171.1.1.0 is directly connected, FastEthernet1/0
172.2.0.0/16 is variably subnetted, 2 subnets, 2 masks
R    172.2.0.0/16 [120/1] via 192.168.1.2, 00:00:01, FastEthernet0/0
O    172.2.2.0/24 [110/2] via 192.168.1.2, 00:00:17, FastEthernet0/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0

Router#
```

(4) 测试 ping:

测试从 p0 到 p2 的连通性。

```
C:\>ping 172.2.2.2

Pinging 172.2.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126
Reply from 172.2.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.2.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

检验发现任意两台之间都是连通的。

思考题:

本题做法结合了任务 1 思考题的思路和任务 3 对 ospf 的配置。配置串口之后，在配置两个路由器的 ospf 即可。

五、 遇到的问题与解决

无问题。